

Ramboll Air Quality Planning Group

Modeling on Google Cloud

GitHub · storage buckets · containers · Google Batch

Modeling in the Cloud Workgroup – May 2026



Ramboll's GCP Approach at a Glance

Our on-site computing center is decommissioning in Q3 2027



COMPUTE

Interactive VMs (cheap) + Google Batch (powerful)

Right-size to the task — no idle clusters



EXECUTABLES

Containers built on Fortran/C compiler images

Same image runs on multiple VM families; no rebuilds necessary



DATA

Google Cloud Storage buckets

Model inputs & outputs; GCS fuse for convenience



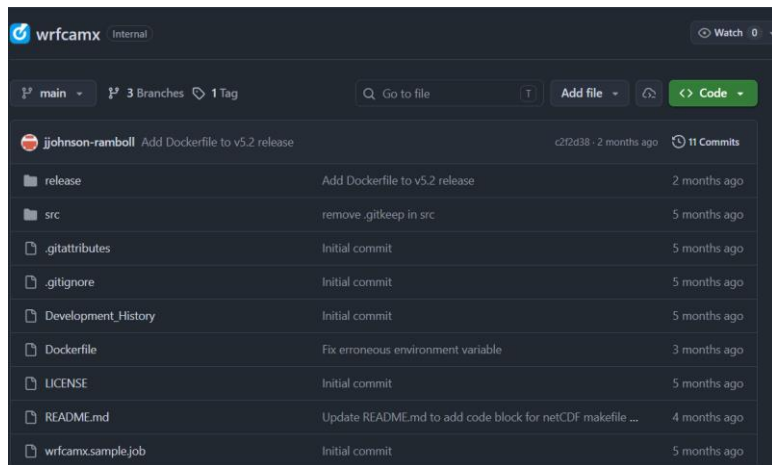
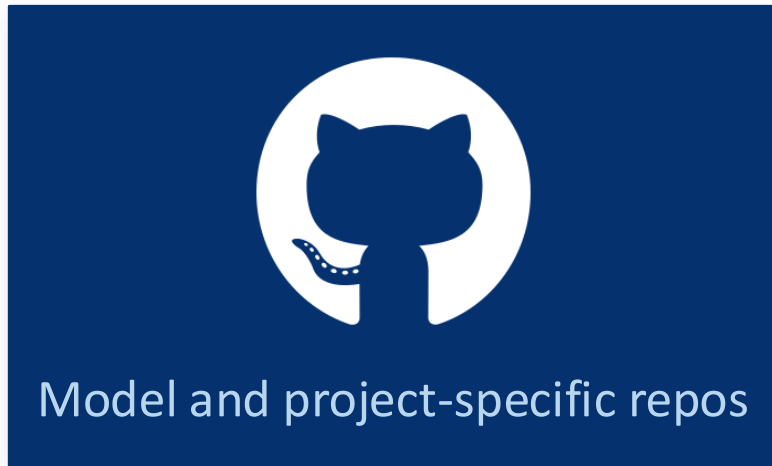
CODE

GitHub repos

Source of code, scripts, and model/processor configuration

GitHub: Single Source of Truth for Code

Codes, scripts and configuration files all live in GitHub



Model & processor source codes

CAMx and processors, WRF evaluation software, etc.



Run scripts & workflow

Shell scripts that drive each model/processor on a VM or Batch job



Configuration files

Namelist, control files, environment definitions

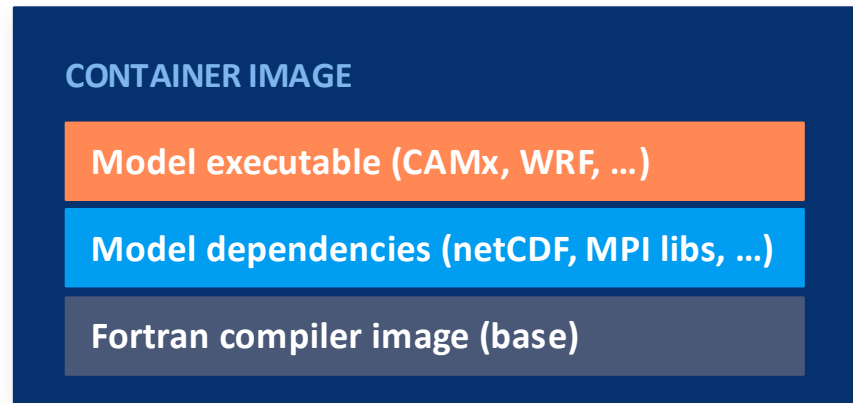


Dockerfiles (build recipes)

How each container image is built — version-controlled with code

Containers: Compile Once, Run (Almost) Anywhere

Built from Fortran compiler images — no manual compiler installs

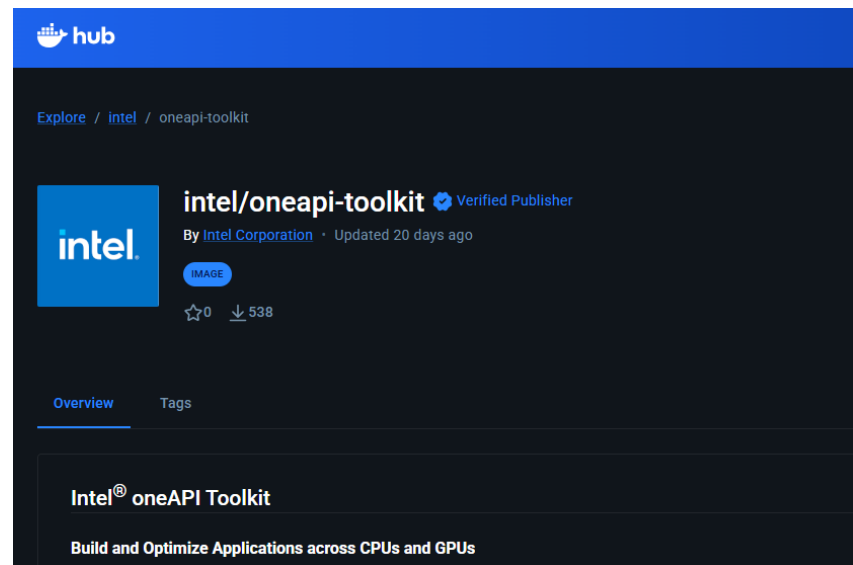


What is it?

Lightweight virtualization technology built on Linux namespaces (think apartments in a building)

Containers

Share kernel and other services with the host
Startup quick with negligible overhead



Build Once

Compiler comes with image, so no need to install or configure one

```
$ docker run intel/oneapi-toolkit:2026.0.0-devel-ubuntu24.04 ifx -V
Intel(R) Fortran Compiler for applications running on Intel(R) 64,
Version 2026.0.0 Build 20260331
Copyright (C) 1985-2026 Intel Corporation. All rights reserved.
```

Portable by Design

Image targets generic x86_64 — same image runs on any Intel/AMD GCP (or another cloud provider's) VM family. But can create images that target chipsets.

Two Compute Modes, Matched to Application



INTERACTIVE VM

Cheap, personal workstations

\$0.14/hr

Persistent disk costs \$90/TB-month

- ✓ Code & script development
- ✓ Pre- and post-processing
- ✓ I/O-heavy data prep, including emissions processing
- ✓ QA/visualize model outputs



N2D-standard-4

General purpose; I/O processing



GOOGLE BATCH

Powerful VMs, spun up per job

\$2.40/hr

Persistent disk and VM torn down when run complete

- ✓ Annual/Seasonal CAMx/WRF simulations divided into segments
- ✓ Each segment runs in parallel



N2D-highcpu-96

CAMx; up to 224 cores available



T2D-standard-X

WRF; vCPUs = physical CPUs

Price/Performance Comparison for 12US2 CAMx Application

GCP general-compute families for "Low" tier: EPA 12US2 domain only (396x246x35), no SAT

INTEL HASWELL

Released 2014

n1-highcpu-96

Older generation

Hourly cost

\$2.93/hr

1-day runtime

42.4 min

1-yr using quarter segments

3.02 days

Power-inefficient — cost inflated relative to performance

INTEL ICE LAKE

Released 2021

n2-highcpu-96

Fastest, but costly

Hourly cost

\$3.38/hr

1-day runtime

29.0 min

1-yr using quarter segments

2.07 days

Only 6% faster than n2d, but +\$0.43/hr

AMD MILAN

Released 2021

n2d-highcpu-96

Default choice

Hourly cost

\$2.95/hr

1-day runtime

30.9 min

1-yr using quarter segments

2.20 days

Best balance of cost and performance

STORAGE COSTS

Bucket storage for 12US2 CAMx project

~\$600

for 6 months of standard storage

Assumes we keep everything; cheaper in practice

+\$600

for 3 more years of "colder" tiers (Autoclass)

Storage Outpaces Compute at Scale

For medium and heavy projects, storage dominates costs

PROJECT TIER	1-YEAR COST (compute vs storage)		Compute	Storage	TOTAL
LOW <i>EPA 12US2 domain only, no SAT</i> 396x246 RUNTIME 30.9 min/day · 2.20 days/qtr	\$524	\$625	compute	storage	\$1,149
MEDIUM <i>36US3 / 12US2 / Colorado 4 km, no SAT</i> 172x148 / 396x246 / 171x126 RUNTIME 41.1 min/day · 2.92 days/qtr	\$698	\$2,069	compute	storage (3.0× compute)	\$2,767
HEAVY <i>Medium + OSAT (16 source groups × 3 regions)</i> RUNTIME 105.3 min/day · 7.50 days/qtr	\$1,788	\$2,910	compute	storage (1.6× compute)	\$4,698

KEY TAKEAWAY

Storage outpaces compute on medium projects (3×) and heavy projects.

Storage assumed: 1 year on Autoclass (progressively colder). Targeted deletion of files substantially reduces these numbers.

Migration Challenges

Changing workflow & habits more difficult than tech implementation



Git & GitHub training

All modelers need to get fluent –
branches, PRs, daily workflow



Storage cost mindset

Shift from "cheap disk forever" to
paying per TB-month



PB-scale migration

Years of archives – what to discard/keep
and upload to buckets



Cost tracking

Attach labels by project, recover from
clients, track overhead costs separately



Remote GCP admin

IAM roles, quotas, networking and more

How This Differs From the AWS HPC Approach

AWS — ACROSS NODES

RAMBOLL — SEGMENTED, PER-VM

Run topology

Single run may span multiple VMs

Run split into segments (e.g. quarters), one VM each

Parallelism

MPI across nodes – tight coupling

MPI within a single VM (96 cores or more); segments independent

Interconnect

EFA / fast cross-node networking

Standard networking – no cross-node interconnect

Shared storage

Lustre / FSx parallel filesystem

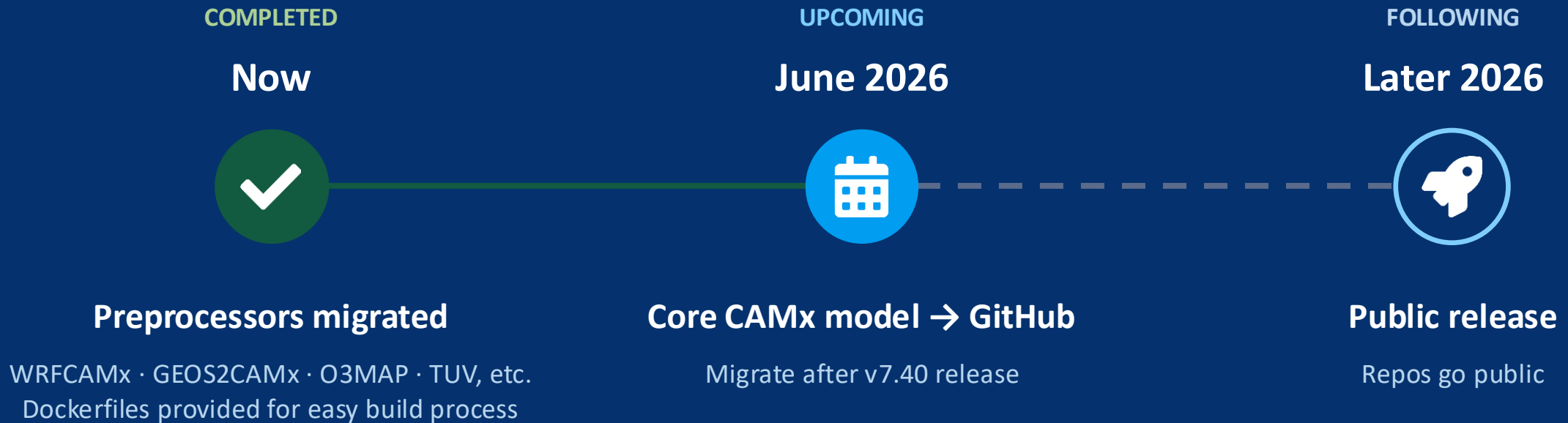
Write directly to GCS buckets with GCS fuse

Executables

Custom AMIs, built per VM type

One container image, (almost) any VM family

CAMx Migration to GitHub



On track for full GitHub migration late 2026.